

Alexander Julian Crabtree

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<https://alexcrabt.github.io/Portfolio/>

Research Summary and Interests

Aspiring researcher with a strong foundation in physics, mathematics, and data science & analytics. Experienced in computational modeling, statistical analysis, and data visualization through both academic coursework, research and professional analytics work. Developed and applied data-driven methods to astrophysical datasets and forecasting models, with an emphasis on reproducibility and automation. Research interests include computational astrophysics, high energy astrophysics, and data-driven simulations.

Education

Johns Hopkins University

Master of Science in Applied Physics (Anticipated Spring 2026)

Relevant Coursework: Electromagnetics, Quantum Mechanics, Space Physics, Plasma Physics, Introduction to Relativity (Planned)

Baltimore, MD

(June 2024 – Present)

Texas A&M University

Bachelor of Science in Mathematics

Minors: Physics, Astrophysics

College Station, TX

(Aug 2017 – May 2021)

Research Experience

Texas A&M Aggienova Research Group

Research Analyst

College Station, TX

(Aug 2020 – Jun 2021)

- Developed Python scripts (BeautifulSoup, Pandas) to extract, process, and analyze astronomical datasets for supernova research.
- Designed Plotly dashboards for visualizing light curves and spectra to support astrophysical data interpretation.
- Improved SOUSA database accessibility using HTML/CSS optimization, enabling faster data retrieval for the research team.
- Contributed to data processing workflows that supported ongoing observational studies in time-domain astrophysics.

Professional Experience

Lone Star Analysis

Promoted through multiple roles

Analytics Engineer

Data Analyst

Associate Data Analyst

Addison, TX (Remote)

(Jun 2022 – Present)

(Jun 2025 – Present)

(Feb 2023 – Jun 2025)

(Jun 2022 – Feb 2023)

- Developed forecasting models using customer data, reducing prediction error by 3%.
- Applied Monte Carlo simulations to capture uncertainty in forecasts and improve accuracy.
- Automated forecasting workflows with Python and SQL, reducing project timelines by up to 2 months.
- Optimized ETL workflows, cutting code size by 90% and reducing processing time to 1 minute.
- Designed interactive Power BI dashboards, enabling stakeholders to make faster, data-driven decisions.
- Standardized Python-based ETL processes and documentation, ensuring scalability and reproducibility.
- Mentored junior analysts in data handling best practices, and in understanding and building upon forecasting models.

Teaching Experience

Education One

Physics and Mathematics Tutor

Frisco, TX
(Sep 2021 – May 2022)

- Provided individualized and small-group instruction in physics, calculus, statistics, and K–12 mathematics to high school and undergraduate students.
- Designed and implemented customized learning plans and problem sets to strengthen conceptual understanding of material.
- Guided students in developing problem-solving strategies for standardized tests (SAT/ACT) and advanced coursework.
- Fostered critical thinking and analytical skills through step-by-step explanations of mathematical and physical principles.

Projects

Plasma Physics Photon Random Walk Simulator

Python

(Aug 2025 – Sep 2025)

- Researched and developed a simulation of a photon's random walk starting from the center of the Sun.
- Implemented physical characteristic functions to calculate density, opacity, temperature, luminosity, and radiative energy transport acting on the photon at various positions.
- Created random angle and adaptive time step functions, incorporating numerical techniques such as the Runge-Kutta method to ensure acceptable error limits.
- Analyzed and visualized simulation data, focusing on the photon's mean free path, anomalous diffusion, and overall displacement.

Swift Supernovae Visualization and Analysis Tools

Python, Plotly, Dash

(Mar 2021 – Jun 2021)

- Programmed and implemented a suite of Python-based tools for visualizing and analyzing Swift Supernovae data from the SOUSA database, focusing on light curve generation and color analysis.
- ***Swift Supernovae UVOT Light Curve Generator:***
Developed Python scripts with Plotly to generate and compare UVOT light curves, supporting time-series analysis of supernova brightness.
- ***Swift Supernovae UVOT Filter Light Curve Generator:***
Automated plotting of UVOT filter light curves across all available data, providing consistent visualizations for comparative study of transient events.
- ***Swift Supernovae Color–Color Diagram Generator:***
Built an interactive Dash application enabling dynamic exploration of supernova color evolution across multiple UVOT filters.
- ***Swift Supernovae All Light Curves App:***
Integrated all prior tools into a comprehensive Dash application that dynamically produces light curves, color light curves, and color–color diagrams based on user input.

Swift Supernovae Data Web Scraper

Python

(Aug 2020 – May 2021)

- Developed Python script to scrape for relevant astronomical data from databases such as NASA/IPAC Extragalactic Database (NED) and the Transient Name Server for the purpose of contributing to SOUSA Supernovae database.

Technical Skills

Programming: Python, SQL, VBA, HTML, CSS

Scientific Computing: Pandas, NumPy, SciPy, Matplotlib, Plotly

Data Visualization: Power BI, Plotly, Matplotlib

Database Management: PostgreSQL, MySQL, Palantir Foundry

Version Control: Git, GitHub

Certificates: IBM Data Analyst Professional